

8) The _____ layer is responsible for moving datagrams from the source host to the destination host.

Answer: network

9) Which of the following is present in both HTTP request line and status line?

Answer: HTTP version number

10) Which of the following protocols run at the application layer? i) SMTP ii) UDP iii) POP iv) DNS

Answer: (i), (iii) and (iv) only

11) The data value to be transmitted is 10100110 and the CRC generator is 11001. What is the CRC value computed?

Answer: 1001

12) Which of the following statement regarding TCP is TRUE?

Answer: TCP provides a connection-oriented service to its applications.

13) Which of the following statements regarding application layer protocol is FALSE?

Answer: A browser can open only one TCP connection at a time.

14) Host A sends 5 data packets... The last ACK packet sent by B has ACK number 99. What is the sequence number of the first data packet sent by A?

Answer: 58

15) Microwave link between satellite and Earth (distance 3.6×10^7 m, speed 2.4×10^8 m/s). Find propagation delay and minimum value of x.

Answer: Propagation delay = 0.15 s, $x = 2.4 \times 10^8$ bits

16) Minimal frame size L in CSMA/CD protocol.

Answer: $2 * d * r$

17) Consider a datagram network using 8-bit IP addresses... For interface 4, give associated range.

Answer: 1010 0000 – 1011 1111, 32

18) Which of the following end-to-end delay component is constant?

Answer: Propagation delay

19) CSMA/CD is a (an) _____ layer protocol.

Answer: link

20) An IP datagram with destination address 192.168.1.10 reaches a router... Which entry will be used?

Answer: 192.168.1.0/24

21) A host uses a variety of protocols... Which statement is FALSE?

Answer: To get an IP address assigned, a host must first discover the IP address of its DHCP server using DNS.

22) Is it necessary for an IP datagram to carry both the source and the destination IP addresses?

Answer: Yes

23) Which of the following is the Web's application layer protocol?

Answer: HTTP

24) How many of the given IP addresses belong to subnet 192.168.160.0/20?

Answer: 2

25) Which of the following statement about 2-dimensional parity bits is FALSE?

Answer: It can correct any two-bit error.

26) Live video streaming often runs over TCP – fast; relatively stable throughput!

Answer: False

27) Which of the following makes up the end-to-end delay?

Answer: processing delay, propagation delay, queueing delay, transmission delay

28) If the fragment offset in an IP header has a value of 100, it means that _____.

Answer: this first byte of payload in the IP datagram is byte 800

29) rdt 4.0 protocol design... Which statement is TRUE?

Answer: Receiver must send ACKs for duplicate packets.

30) The _____ field in IP header prevents an IP datagram from continuously wandering through the Internet.

Answer: TTL

31) The access method for wireless LAN (IEEE 802.11) is based on _____.

Answer: CSMA/CA

32) Which of the following best characterizes CSMA/CD protocol?

Answer: Listen to the channel before accessing it.

33) Can HTTP response message contain an empty body?

Answer: Yes, if the requested object has been moved to a new location.

34) In circuit switching, circuit segment is idle if not used by call.

Answer: Agree

35) Troubleshooting web access: dig, ping, traceroute, telnet. Which one checks if www.example.org is listening on port 80?

Answer: You run telnet to check if www.example.org is listening on port 80.

36) Which of the following statements about TCP/UDP checksum are TRUE?

Answer: (iii) and (v) only

37) Which of the following fields is present in both UDP header and TCP header?

Answer: Checksum

38) Ethernet uses a _____ physical address that is imprinted on the NIC.

Answer: 48-byte

39) For the screenshot, what is the IP address of the hand phone in CIDR format?

Answer: 117.91.224.0/22

40) In a host sending data, UDP receives data from the _____ layer and sends to the _____ layer.

Answer: application; transport

41) Ethernet provides an unreliable service. Therefore, _____.

Answer: Ethernet drops a frame that fails error checking without retransmission

42) The first line in an HTTP request message is called the _____ line. The first line in an HTTP response message is called the _____ line.

Answer: request; status;

43) Which of the following technique is most essential in reliable transmission protocols?

Answer: Encapsulation

44) The _____ layer is responsible for moving frames from one hop (node) to the next.

Answer: link

45) The Internet protocol stack consists of _____ protocol layers.

Answer: 5

46) 1s complement is used as checksum in _____.

Answer: Both TCP and UDP

47) In Non-persistent HTTP, the client closes the TCP connection after fetching one object.

Answer: True

48) What is the 1s complement of the sum of 11011010 and 01100101?

Answer: 00111110

49) FTP runs over UDP – need to ensure files are uploaded/downloaded intact.

Answer: False

50) Which of the following is a VALID subnet mask?

Answer: 255.255.255.128

51) Which of the following statements regarding the Internet is TRUE?

Answer: The Internet is structured as a network of networks.

52) What does HTTP stand for?

Answer: Hypertext Transfer Protocol

53) Which of the following layers doesn't appear in the Internet Protocol Stack?

Answer: session

54) Huge file over TCP connection... How many TCP segments are used?

Answer: 4,194,295

55) The _____ layer of the Internet protocol stack is responsible for delivering data from sending process to receiving process.

Answer: transport

56) Two hosts 2000 km apart... How long does it take for B to receive all the packets?

Answer: 0.048 seconds